Customer Churn Analysis and Retention Strategy

Executive Summary

Customer churn poses a significant challenge for subscription-based businesses, directly impacting revenue and customer lifetime value. This project develops a machine learning-based churn prediction framework to identify customers at risk of leaving and recommend targeted retention strategies.

Using telecom customer data, exploratory data analysis, feature engineering, and predictive modeling techniques were applied to understand churn behavior and improve customer retention decision-making.

Business Problem

Customer acquisition is significantly more expensive than customer retention. Identifying customers likely to churn enables organizations to proactively intervene and reduce revenue loss.

Objectives

- Identify key factors influencing customer churn.
- Develop predictive models for churn classification.
- Segment customers based on churn risk.
- Recommend actionable retention strategies.

Dataset Overview

- Industry: Telecommunications
- Records: 7,000+ customers
- Target Variable: Churn (Yes/No)

Key Features

- Customer tenure
- Contract type
- Monthly charges
- Total charges
- Internet service
- Payment method

Exploratory Data Analysis

Key Findings

1. Customers with month-to-month contracts exhibited significantly higher churn rates.
2. Customers with shorter tenure were more likely to leave.
3. Higher monthly charges correlated with increased churn probability.
4. Long-term contract customers demonstrated stronger retention.

Model Development

Algorithms Evaluated

Model	Purpose
Logistic Regression	Baseline classification
Random Forest	Ensemble learning
XGBoost	Gradient boosting
LightGBM	Optimized gradient boosting

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

Best Performing Model

LightGBM achieved the strongest overall performance with ROC-AUC approximately 0.89.

Key Churn Drivers

High Impact Variables

1. Contract Type
2. Tenure
3. Monthly Charges
4. Internet Service Type
5. Payment Method

Customer Segmentation

High Risk

- Short tenure
- Month-to-month contracts
- High monthly charges

Medium Risk

- Moderate tenure
- Mixed service usage

Low Risk

- Long-term contracts
- High engagement levels

Business Recommendations

1. Contract Conversion

Offer incentives for customers to migrate from month-to-month plans to annual contracts.

2. Targeted Retention Campaigns

Identify high-risk customers and provide personalized discounts or loyalty rewards.

3. Customer Engagement Programs

Improve customer experience through proactive support and service improvements.

4. Risk Monitoring Dashboard

Deploy a dashboard to continuously monitor churn probability and retention performance.

Business Impact

- Improved customer retention decision-making.
- Early identification of high-risk customers.
- Reduced potential revenue loss.
- Enhanced customer lifetime value management.

Conclusion

The churn prediction framework successfully identifies customers at risk of leaving and provides actionable business insights. The combination of predictive analytics and retention strategies can help organizations reduce churn and improve long-term profitability.